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THE PETROLOGY OF LUNAR BRECCIA 15445 AND PETROGENETIC IMPLICATIONS[1,2]

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ABSTRACT

Lunar breccia 15445, collected near the rim of Spur Crater, contains iron-rich olivine and orthopyroxene mineral clasts and two types of white lithic clasts. The dominant type (Type A) is composed principally of anorthite — magnesian orthopyroxene and appears to be a low-pressure cumulate from a magnesian, high-Al basalt. A minor type (Type B) is composed of olivine + chrome pleonaste + aluminous orthopyroxene + minor plagioclase. Type B clasts have affinities to terrestrial spinel peridotites and have a REE pattern unique for the moon. Considerations of the observed mineralogy and the trace element geochemistry suggest that garnet was originally a phase in this peridotite but was subsequently lost during subsolidus reequilibration. Hence, Type B clasts provide indirect evidence for the existence of garnet in the lunar interior.

Data are presented here on breccia 15445, collected during the Apollo 15 mission on the rim of Spur Crater at station 7. This sample, probably derived from the deepest stratigraphic level ($\sim$20m) of Spur Crater (ALGIT 1972), is a breccia with abundant white clasts in a dark matrix. Clasts composed principally of plagioclase and orthopyroxene are abundant (Type A); whereas a minor clast type (Type B) contains conspicuous, pink, chrome-pleonaste + olivine + plagioclase + orthopyroxene.

The major and trace element geochemistry and compositions of the major mineral phases in these two clast types and the breccia matrix are described below and the implications of these data to the problem of the constitution of the Apennine Front (pre-Imbrian crust) are briefly discussed.

PETROCHEMISTRY OF MATRIX AND CLASTS

The breccia matrix is composed of interlocking anhedra of olivine, plagioclase, and intergranular ilmenite. Major and trace element data are given in table 1. The composition of the matrix contrasts with all of the Front soils (LSPET 1972, table 2)—higher in MgO and lower in FeO and CaO. The Mg/Mg + Fe (atomic) ratio (0.74) is higher than the most magnesian Front soil, 15301 (table 2) and the Apollo 15 green glass. The percentage of Al_2O_3 is higher than any of the surface soils from Spur Crater. In contrast, the normalized REE pattern (fig. 1) is very similar to the bulk surface soil at Spur Crater, although the Eu anomaly in 15445 is smaller.

The matrix encloses individual mineral clasts (distinct from the larger lithic clasts) of unbrecciated plagioclase, recrystallized olivine, and rare orthopyroxene. Two types of olivine are present, a magnesian type (Fo 83) and a more iron-rich type (Fo 63). The orthopyroxene mineral clasts are also iron-rich (Mg/Mg + Fe = 66–68) and chemically distinct from the more magnesian orthopyroxenes in the lithic clasts.

Of the white lithic clasts, 95% are one distinct type, characterized by extreme brecciation, no pyrometamorphism, and an essentially simple plagioclase + orthopyroxene mineralogy. Major element data (table 1) indicate these clasts resemble the "anorthositic norites and troctolites" suite

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TABLE 1

Chemical Data for Black and White Breccia 15445 and Two Coarse Fine Fragments from the Same Sampling Station

	15445,25 Black Matrix (57.7 mg)		15445,17 White Matrix (51.0 mg)		15445,71 Spinel-rich Clast (55.7 mg)	15303,G2 (34.9 mg)		15303,3,7 (23.2 mg)
SiO_2 (%)	*(44.6)*	—	*(48.7)*	—	—	*(47.25)*	—	—
TiO_2 (%)	*(1.47)*	1.49	*(0.15)*	0.140	0.588	*(1.18)*	1.21	0.826
Al_2O_3 (%)	*(16.66)*	16.2	*(23.76)*	20.8	7.2*	*(16.88)*	16.3	—
FeO (%)	*(9.83)*	10.2	*(3.88)*	3.8	—	*(11.24)*	13.3	—
MnO (%)	*(0.14)*	—	*(0.08)*	—	—	—	—	—
MgO (%)	*(16.0)*	15.5	*(9.94)*	9.7	31.1*	*(11.20)*	11.0	7.8
CaO (%)	*(10.04)*	9.6	*(13.26)*	12.6	1.9	*(10.82)*	11.1	14.8
Na_2O (%)	—	0.55	—	0.33	0.10	*(0.40)*	0.45	0.46
P_2O_5 (%)	*(0.21)*	—	*(0.03)*	—	—	—	—	—
S (%)	*(0.06)*	—	*(0.00)*	—	—	—	—	—
K (ppm)	—	1,370	—	582	—	*(14.90)*	1,500	558
Cr (ppm)	—	1,750	—	1,560	—		2,310	—
Li (ppm)		14.1		—	—		12.1	—
Rb (ppm)		3.56		1.43	—		4.39	1.23
Sr (ppm)		160		130	—		—	170
Ba (ppm)		237		61.9	25.0		254	72.3
La		22.1		4.02	1.84		21.8	4.24
Ce		57.6		11.1	3.2		57.2	11.9
Nd		35.7		5.91	2.89		34.8	7.25
Sm		10.1		1.65	1.05		9.95	1.97
Eu		1.85		0.929	0.196		1.30	1.09
Gd		11.9		2.05	2.09		13.1	2.61
Dy		13.2		2.69	4.88		13.4	2.88
Er		7.71		1.72	4.04		7.98	1.96
Yb		6.90		1.78	3.89		7.30	1.82
Lu		1.02		0.268	0.573		1.06	0.267

Note.—Data in italics are from the same XRF technique and standardization used for the Apollo 15 PET analyses, except for 15303,G2 which is an average of four microprobe analyses corrected for deadtime, drift, background, and matrix effects.

* These data include corrections for undissolved pink spinel; other concentrations may be up to 5% high.

TABLE 2

Major Element Compositions of Some Rock Types from Apollo 16, 16 Luna-20 Sites, for Comparison with Table 1

	1	2	3	4	5	6	7
SiO_2	45.91	45.43	46.25	46.94	46.56	46.09	37.5
TiO_2	1.17	0.42	0.95	1.40	1.25	0.79	0.05
Al_2O_3	14.53	7.72	17.74	16.71	18.83	20.01	15.9
FeO	14.05	19.61	10.97	11.18	9.67	9.36	5.8
MnO	0.19	n.d.	0.17	0.15	0.20	n.d.	0.16
MgO	12.12	17.49	11.10	9.95	11.04	10.27	33.7
CaO	10.70	8.34	11.79	11.19	11.60	12.60	6.2
Na_2O	0.35	0.12	0.42	0.51	0.37	0.37	0.14
K_2O	0.16	0.01	0.15	0.25	0.12	0.08	0.04
P_2O_5	0.15	n.d.	0.11	0.25	n.d.	n.d.	0.02
S	0.04	n.d.	0.05	0.08	n.d.	n.d.	n.d.
Mg/Mg + Fe	0.60	0.61	0.64	0.61	0.67	0.66	0.91

Note.—1, Bulk soil 15301 from station 6 (analyzed by M. Rhodes, LSPET 1972); 2, average analyzed green glass (Ridley, Reid et al. 1972); 3, matrix 15445,98) of breccia 15459 from station 6 (Rhodes, analyst); 4, bulk composition of breccia 15265 from station 6 (LSPET 1972); 5, average low-K, Fra Mauro basalt glass from Apollo 15 Front soils (Reid, Ridley et al. 1972); 6, average low-K Fra Mauro basalt glass from Luna-20 soil (Warner et al. 1972); 7, spinel troctolite clast 67435,14 (Prinz et al. 1973*a*).

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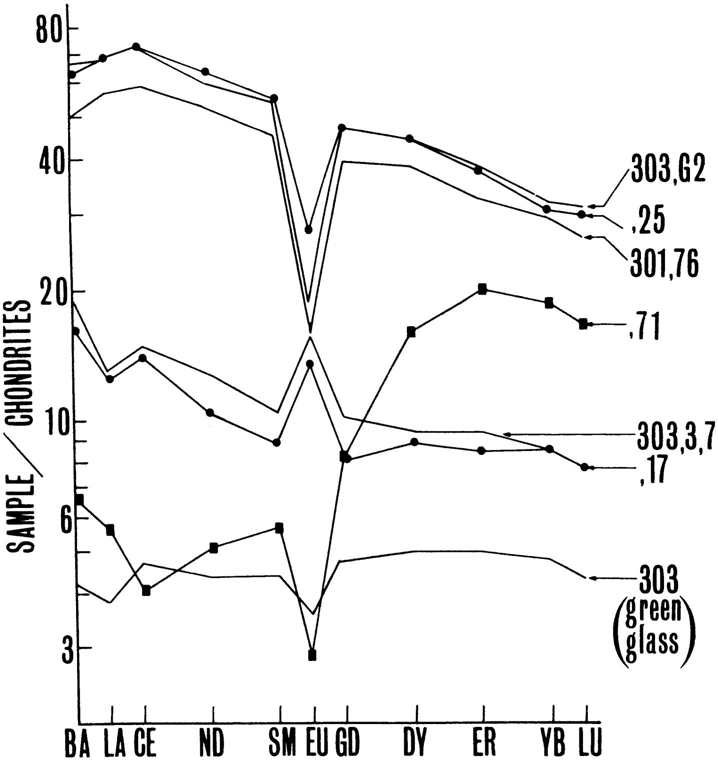

FIG. 1.—Normalized REE patterns for components of breccia 15445: matrix (,25); white "anorthositic" clast Type A (,17); ultramafic clast spinel Type B (,71). For comparison are also shown; high-alumina basaltic glass from Spur crater soil 15303 (303,G2); bulk soil 15301 from Spur crater (301,76); plagioclase-vitrophyre fragment from Spur crater soil 15303 (15303,3,7); green glass "clod" from Spur crater (303).

described from the Luna 20 soil (Prinz et al. 1973*b*), and "highland basalt" glasses (Reid, Ridley et al. 1972). The normalized REE pattern (fig. 1) shows a moderate enrichment of light over heavy REE and a conspicuous positive Eu anomaly. A similar pattern has been observed for a plagioclase vitrophyre (15303,3,7), also collected at Spur Crater.

Plagioclase in Type A clasts is extremely anorthitic (An 96), intensely crushed and sheared, and commonly displays a "snowball" texture commonly observed in some feldspathic Apollo 16 breccias (LSPET 1973). It is distinct from the less calcic KREEP plagioclases (Hubbard, Gast, Meyer et al. 1972; Ridley, Brett et al. 1972) but similar to calcic plagioclases from coarse anorthositic soil fragments at Spur Crater (Phinney et al. 1972; Cameron et al. 1973) and from the ANT suite of Prinz et al. (1973*b*). Associated orthopyroxenes are aluminous bronzites (table 3) distinctly more magnesian than the average orthopyroxene in "anorthositic" norites and troctolites from Luna-20 soil (Prinz et al. 1973*b*). Type A clasts also frequently contain minor magnesian ilmenite (MgO $\lesssim$ 7.8 wt %) and rarely contain Cr-Zr-armalcolite (TiO_2, 64.2; FeO, 8.5; Cr_2O_3, 11.3; Al_2O_3, 1.6; MgO, 2.1; CaO, 4.0; ZrO_2, 6.4), that extends the range of chromium solid solution in Zr-armalcolite as discussed by Haggerty (1972).

Major and trace element data for Type B clasts are shown in table 1 (15445,71).

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TABLE 3

MICROPROBE ANALYSES OF ORTHOPYROXENES IN CLASTS AND MATRIX OF 15445, AND RELEVANT APOLLO SAMPLES

	1	2	3	4	5	6	7	8	9	10	11	12	13
SiO_2	...	...	...	56.51	54.04	52.80	54.29	54.22	53.90	54.90	52.71	55.0	53.51
Al_2O_3	...	...	...	2.44	1.52	1.98	0.86	1.99	0.92	3.00	0.28	1.2	0.66
TiO_2	...	...	...	0.15	0.17	0.13	0.50	0.14	0.53	0.55	0.17	0.7	0.35
CaO	2.17	1.54	1.21	<0.01	1.25	1.31	1.64	1.15	1.56	1.89	1.33	1.5	2.61
FeO	19.48	20.79	20.58	5.70	10.77	9.46	11.81	9.78	12.96	11.96	24.18	10.5	16.69
MgO	23.07	23.24	23.58	35.67	30.92	31.67	29.78	32.19	28.91	28.46	20.36	30.0	24.26
Cr_2O_3	...	...	...	...	0.53	0.57	0.42	0.61	0.45	...	0.11	0.4	0.74
Total	...	...	...	100.47	99.20	97.92	99.30	100.08	99.23	99.90	99.7	99.3	99.11
Mg/Mg + Fe .	0.68	0.66	0.66	0.92	0.83	0.86	0.82	0.85	0.80	0.81	0.60	0.83	0.72

NOTE.—1–3, Partial analyses of orthopyroxene single-crystal clasts in matrix of 15445; 4, partly crushed orthopyroxene in 15445,9; some areas contain up to 3.4 wt% Al_2O_3; orthopyroxene in close relationship with plagioclase + olivine + pleonaste; 5, average of three analyses from 15445,63A, a plagioclase-orthopyroxene-armalcolite clast; 6,7, most and least aluminous pyroxenes respectively in a plagioclase-rich clast 15445,63B; 8,9, most and least aluminous pyroxenes in plagioclase-rich clast 15445,61; 10, aluminous bronzite in KREEP basalt 14310 (Ridley, Brett et al. 1972); 11, orthopyroxene in anorthosite gabbro 15415 (Hargraves and Hollister 1972; total includes 0.54% MnO); 12, most magnesian orthopyroxene in 12033 "KREEP" breccia (Meyer et al. 1971); 13, orthopyroxene core to pyroxene in 12065 (Hollister et al. 1971: analysis includes 0.29% MnO).

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Precise major element data are precluded by the small sample available for analysis. Nonetheless, it is clear from the high MgO, and low FeO and Al_2O_3 that the Type B clasts are ultrabasic. The closest similarity appears to be with a spinel troctolite at the Apollo 16 site (Prinz et al. 1973*a*, table 2). In addition, the REE pattern is distinctive, showing a considerable enrichment of heavy over light REE, a large negative Eu anomaly, a maximum at Er and a break in slope at Gd that produces a shallower Ce/Sm slope; La, Ba are enriched relative to Ce and Nd. Such extreme, relative enrichment in heavy REE is unknown among lunar materials; except for Apollo 11 low-K basalts, such materials show light REE enrichment (Gast et al. 1970; Hubbard and Gast 1971; Hubbard, Gast, Rhodes et al. 1972; Hubbard, Nyquist et al. 1972).

Anderson (1973) has described in detail the mineralogy of a Type B fragment 15445,10. Data from 15445,60 and 15445,9 are essentially similar. Olivine, chrome-pleonaste, and orthopyroxene are in grain contact, but their relationship to sparse fine-grained areas of calcic plagioclase remains unclear. Coarse-grained zones result from the preservation of unbrecciated olivine and spinel; from this scant evidence it is surmised that Type B clasts may originally have had a coarse-plutonic texture. The clasts subsequently underwent brecciation, shearing (but not pyrometamorphism), and incorporation in the breccia.

The ferromagnesian minerals are all highly magnesian. Olivine (Fo 91–88) coexists with orthopyroxene (Mg/Mg + Fe = 0.92) and chrome-pleonaste (Mg/Mg + Fe = 0.81). Minor plagioclase is also extremely anorthitic (An 98). Orthopyroxenes with a combination of very high Mg/Mg + Fe and high Al_2O_3 in such an ultrabasic assemblage have not previously been described from the moon (table 3). They cannot be related simply to pyroxenes in KREEP basalts (Meyer et al. 1971; Ridley, Brett et al. 1972), anorthositic rocks (Hargraves and Hollister 1972; Cameron et al. 1973; Phinney et al. 1972; Prinz et al. 1973*b*), or to primitive orthopyroxenes in mare basalts (Hollister et al. 1971). The combination of high Mg/Mg + Fe and high Cr_2O_3 in the spinel is also unusual. Rare pleonastes high in $MgAl_2O_4$ have been observed from Apollo 12 (Keil et al 1971), Luna-16 soil (Jakeš et al. 1972), Apollo 14 (Mason et al. 1972) and are the dominant spinel type at the Luna-20 site (Reid et al. 1973; Brett et al. 1973). (See tables 4 and 5 for comparisons of analyses of lunar pleonastes and olivine, respectively.) All these examples, however, have little chromite solid solution, except rare Luna-20 pleonastes (Brett et al. 1973) which have lower Mg/Mg + Fe ratios than those described here.

DISCUSSION

Matrix.—The distinctive chemistry of the matrix—in particular, the high Al_2O_5, MgO, and Mg/Mg + Fe—is difficult to interpret in terms of mixing of the most abundant components in the Front soils (Reid, Warner et al. 1972). Matrices of other Spur Crater breccias (table 2) are also unlike the Front soils in major element chemistry, and some are also distinct in major element composition from the matrix of 15445. The latter cannot be considered simply a strongly indurated soil, and appears to represent a chemically distinct unit in a regolith that shows gross stratigraphic inhomogeneities. The slope of the REE, Ba abundance pattern (fig. 1) is typical of such non-mare materials as KREEP (Hubbard, Gast, Rhodes et al. 1972), VHA basalts (Bansal et al. 1973) and brown glass matrix breccias (Hubbard et al., unpublished data), further suggesting that this breccia is another pre-Imbrian material. The combination of high Al_2O_3, MgO, Mg/Mg + Fe, and small Eu anomaly for the overall REE concentration distinguishes this material from other Apennine Front samples.

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TABLE 4

MICROPROBE ANALYSIS OF CHROMIUM-BEARING PLEONASTE FROM 15445,9 COMPARED WITH OTHER LUNAR PLEONASTES

	1	2	3	4	5	6	7	8	9	10
Cr_2O_3	13.40	13.23	13.72	11.84	2.16	5.80	7.4	8.0	1.24	2.04
FeO	9.52	9.54	9.59	9.24	3.33	7.34	8.6	8.0	5.55	5.05
MgO	20.10	20.25	20.26	20.71	25.9	21.11	22.1	24.0	23.54	25.35
Al_2O_3	56.97	57.41	57.32	59.03	68.0	64.75	61.2	59.0	66.79	67.82
TiO_2	0.01	0.00	0.02	0.00	n.d.	0.02	0.3	n.d.	0.05	0.08
MnO	0.06	0.06	0.06	0.06	n.d.	0.04	n.d.	n.d.	n.d.	0.04
Total	100.06	100.49	100.97	100.88	99.39	99.06	101.6	99.0	97.87	100.42
Mg/Mg + Fe	0.80	0.80	0.81	0.81	0.94	0.84	82.0	0.84	89.0	90.0
Cr/Cr + Al	0.13	0.13	0.14	0.12	0.021	0.057	0.075	0.08	0.02	0.02

NOTE.—1–4, Pleonaste in 15445,9; 5, pleonaste associated with olivine + plagioclase in anorthosite clast in 10019,22 (Keil et al. 1970); 6, pleonaste in Luna 16 fines (Jakeš et al. 1972; note that the FeO value of 2.54 wt% [p. 259] should read 7.54 wt%); 7, pleonaste in 14163,42 (Mason et al. 1972); 8, chrome-pleonaste associated with plagioclase (An 94) and olivine (Fo 85) (approximate analysis in Steele 1972); 9, chrome-pleonaste associated with plagioclase, olivine (Fo 80), and traces of low-Ca pyroxene, in 15602 (J. B. Reid 1972); and 10, high-Mag spinel from Luna 20 soil (Reid et al 1973).

TABLE 5

MICROPROBE ANALYSES OF OLIVINES IN 15445 AND RELATED LUNAR SAMPLES

	1	2	3	4	5	6	7
SiO_2	40.85	40.17	38.73	36.24	38.9	38.9	41.2
FeO	8.55	11.17	16.40	32.63	22.9	17.8	7.5
MgO	50.20	48.23	44.43	30.73	38.8	42.4	51.3
CaO	<0.01	0.06	0.13	0.27	0.06	0.25	0.24
TiO_2	<0.01	0.01	0.02	<0.01	0.07	0.03	0.03
Al_2O_3	0.22	0.15	0.36	0.38	0.15	n.d.	0.38
Total	99.83	99.79	100.07	100.25	100.94	99.65	100.74
Mg/Mg + Fe	0.91	0.88	0.83	0.63	0.75	0.81	0.92

NOTE.—1–3, Variations in olivine compositions in 15445,60; the majority of analyzed olivines have the composition of olivine no. 1; 4, anhedral olivine single-crystal clasts in matrix of 15445; 5, olivine associated with pleonaste + plagioclase in 14319,11a-3 (analysis includes 0.03% Cr_2O_3, 0.01% NiO; Roedder and Wieblen 1972); 6, olivine associated with pleonaste + plagioclase in anorthosite clast in 10019,22 (analysis includes 0.04% Cr_2O_3; Keil et al 1970); 7, olivine associated with pleonaste + plagioclase in troctolite 67435,14 (Prinz et al. 1973*a*).

Clast Type A.—There is a measure of agreement that some highly aluminous lunar rocks may be cumulates—especially those whose major and trace element compositions deviate substantially from those of lunar basalts. Although the Type A clasts are strongly sheared, textural evidence indicates that original feldspar-rich and orthopyroxene-rich bands have been preserved. Such a texture may indicate an original layering in the rock and hence support a cumulate origin for Type A clasts. In addition, the substantial positive Eu anomaly contrasts markedly with the negative Eu anomalies associated with known high-alumina lunar basalts but is consistent with some degree of plagioclase accumulation.

If we make the reasonable assumption that minerals in Type A clasts were originally equilibrated with a basaltic melt, the distribution coefficients for major elements (Roeder and Emslie 1970; Medaris 1969) should provide some restraints on the melt composition. Figure 2 indicates the Mg/Mg + Fe ratio of melt in equilibrium with Type A orthopyroxenes. For comparison, the range in Mg/Mg + Fe

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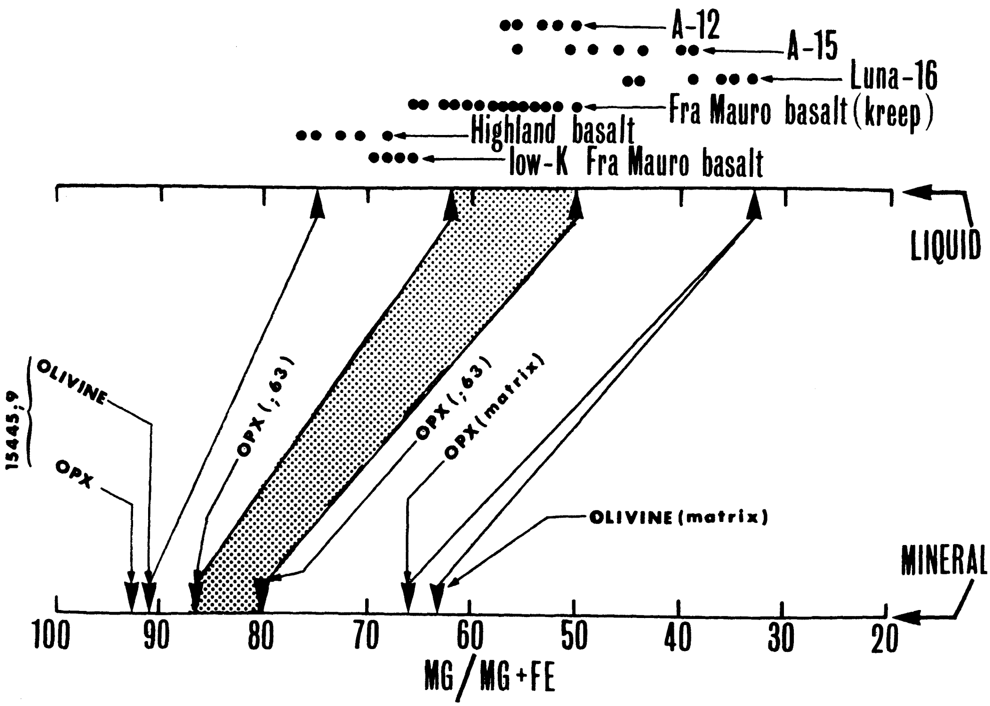

FIG. 2.—Mg/Mg + Fe ratio for olivines and orthopyroxenes from 15445, and corresponding ratios of basaltic liquids in equilibrium with these minerals. Distribution coefficient data from Roeder and Emslie (1970) and Medaris (1969). The range of Mg/Mg + Fe ratios for known, or hypothesized lunar basaltic liquids are taken from analysis of Apollo 12 samples by LSPET (1970) and Compston et al. (1970); of Luna 16 samples by Jakeš et al. (1972), Vinagradov (1971), Grieve, McKay, and Weill (1972), and Steele and Smith (1972); of Apollo 15 samples by Reid, Warner et al. (1972) and LSPET (1972); of Fra Mauro basalt (KREEP) samples by Hubbard, Gast, Rhodes et al. (1972), Anderson and Smith (1971), Keil et al. (1971), Meyer et al. (1971), and Reid et al. (1972*c*); of low-K Fra Mauro basalt samples by LSPET (1973), Reid, Warner et al. (1972), and Warner et al. (1972); and of Highland basalt samples by Reid, Ridley et al. (1972).

ratios of mare basalts, two types of high-alumina basalt, and Highland basalt (Reid, Ridley et al. 1972) are also shown. The melt Mg/Mg + Fe values do not define one specific parent, although the close association of orthopyroxene and plagioclase eliminates a mare basalt parent. Plagioclase and orthopyroxene would be the first phases to separate from a KREEP magma near the critical plane of oversaturation. However, the liquidus KREEP plagioclase is less calcic (An 90–92, Ringwood et al. 1972) compared with the consistently more calcic (An 96) variety in type A clasts.

Crystallization from an aluminous melt with high Mg/Mg + Fe ratio, but low total alkalies would be consistent with the presence of calcic plagioclase, magnesian orthopyroxene, magnesian ilmenite, and rare Zr-armalcolite. Low-K Fra Mauro basalt (Reid, Warner et al. 1972, table 2), high-alumina basalt (Prinz et al. 1973*b*), and very high-alumina basalt (Bansal et al. 1973) are all possible parental materials for Type A minerals.

We also note that the lithic clast orthopyroxenes are distinct from those occurring as single mineral clasts in the matrix. This latter type is closely related

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to the iron-rich olivine mineral clasts, and both minerals could have crystallized from the same iron-rich melt (fig. 2).

Petrogenesis of Type B clasts.—The overall mineralogy of these clasts is that of spinel peridotite, except clinopyroxene is absent. The mineral assemblage olivine + chrome pleonaste + orthopyroxene + plagioclase could not produce the observed REE pattern, if these minerals were in equilibrium with any known lunar basaltic melt (Philpotts et al. 1972). The REE pattern is similar, however, to that of pyropic garnet in terrestrial peridotites (Philpotts et al. 1972). The Eu anomaly is uniquely lunar. We conclude that the REE pattern may record the prior existence of garnet in Type B clasts, the pattern being modified for Ba, La, Ce, and Nd by the presence of one or more additional components.

The proposed petrogenesis relies upon analogy with terrestrial garnet-bearing peridotites. Experimental studies indicate that with decreasing pressure peridotites pass from the stability field of garnet into that of aluminous spinel; Reid and Dawson (1972) have described a suite of terrestrial garnet peridotites that have preserved mineralogical evidence for the isochemical reequilibration of garnet, producing spinel + orthopyroxene + minor clinopyroxene. The ultramafic lunar xenoliths described here may also have undergone subsolidus reequilibration during ascent, in which the original garnet broke down to give orthopyroxene + pleonaste. By inference, the original ultramafic sample must have been essentially an olivine + garnet rock. Significant grossularite-andradite solid solution in the original garnet would require the presence of diopside in the subsolidus assemblage, but the very low Ca content of the orthopyroxene (lower than a previously described lunar orthopyroxene; Anderson 1973) suggests that the system was depleted in calcium. Such a situation might result if the assemblage garnet + olivine was a residue following partial melting. Under such conditions, the olivine would have equilibrated with a melt having Mg/Mg + Fe ratio of about 0.75 (fig. 2). Data from Reid and Dawson (1972) indicate that reequilibration produces a spinel with a lower Mg/Mg + Fe ratio and an orthopyroxene with a higher Mg/Mg + Fe ratio than the original garnet. By analogy, the original lunar garnet would have an Mg/Mg + Fe ratio approaching 0.85. The combination of high Al_2O_3, Cr_2O_3, and Mg/Mg + Fe ratio in the pleonastes also suggests they did not form by crystallization from a chrome-poor, high-alumina melt—a proposal currently popular for the origin of most Apollo 14, 16, and Luna-20 pleonastes (Roedder and Weiblen 1972, Prinz et al. 1973*b*, Brett et al. 1973). Possibly, subsolidus formation of pleonaste from a chrome-rich garnet could produce the observed chemical features, although intense shearing of the Type B clasts precludes the preservation of textural evidence that might confirm such a speculation.

Terrestrial garnet peridotite xenoliths invariably show evidence of equilibration with a silicate melt (Philpotts et al 1972), and many show evidence of "contamination" by the host magma that propagated them to the surface. Modification of the REE pattern in the lunar ultramafic xenoliths may also suggest some contamination of the simple mineral assemblage of olivine and garnet by interstitial melt. The large Eu anomaly could be accounted for if this was a primitive feature of lunar ultrabasic assemblages, or if the garnet had equilibrated with a melt that itself had a large negative Eu anomaly. Significant Eu anomalies are produced by equilibration of garnet with KREEP basalt melt, very high Al_2O_3 basalts, or with Apollo 11 basalts, but not by equilibration with other mare basalts. However, this type of equilibration does not produce the flattening of the Ce-Sm portion of the REE pattern.

A further hypothesis—compatible with

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similar observations amongst terrestrial peridotites—is that the garnet-bearing assemblage, originally in equilibrium with a melt enriched in light rare earths and with a large Eu anomaly, retained some intergranular fluid during ascent to the surface. Dark reaction rims enriched in Ti, Fe and containing a small amount of anorthitic plagioclase around chrome-pleonaste may be indications of low-pressure reaction of the ultramafic minerals with intergranular fluid.

CONCLUSIONS

In the absence of extensive pyrometamorphism, minerals in lithic clasts and single-crystal clasts in breccias provide useful information on the chemistry of basaltic liquids with which they may have equilibrated.

Single-mineral clasts in breccia 15445 could only have equilibrated with an iron-rich magma (the presence of orthopyroxene suggests the magma was unlike mare basalts), and had probably already undergone extensive fractionation. Possibly, such minerals are related to the iron-rich ferromagnesian minerals reported from some anorthositic rocks (Phinney et al. 1972; Cameron et al. 1973; James 1972).

Anorthositic norite clasts (Type A) are principally composed of anorthitic plagioclase and bronzite with minor magnesian ilmenite and Cr-Zr armalcolite. Equilibration of the individual minerals with a high-alumina, magnesian basaltic magma is suggested, probably more depleted in alkalies than KREEP basalts (table 2). On a broader scale, minerals similar to those reported here occur in some Apollo 15 anorthositic rocks (Phinney et al. 1972; Cameron et al. 1973) and in the Luna-20 sample (Reid et al. 1973; Prinz et al. 1973*b*), suggesting a low-K, magnesian basalt parent for these rocks also.

Rare ultramafic clasts (Type B) in 15445 show a REE pattern compatible with an early history involving garnet. The trace element and mineral data indicate subsolidus reequilibration within the aluminous spinel field, and possibly retention of some intergranular melt. This ultramafic rock provides the first evidence for garnet in the lunar interior.

The evidence remains indirect, but compelling, that a suite of high-alumina magnesian basalt magmas, less enriched in alkalies than KREEP, have produced a range of aluminous rocks—including the low-K calcic anorthosites (Hubbard and Gast 1971), and the magnesian olivines, orthopyroxenes, and ilmenite found in some anorthositic clasts. From different lines of approach, Hubbard, Gast, Meyer et al. (1972), Reid, Warner et al (1972) and Prinz et al. (1973*b*), as well as the present study arrive at essentially this conclusion.

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